

Figure 2 shows the boot chart when a boot chart *png* file is viewed.

However, a boot chart is not necessary for Ubuntu versions later than 15.10. To get very brief information regarding boot up speed, use the following command:

```
systemd-analyze
```

Figure 3 shows the output of the command *systemd-analyze*.

The command *systemd-analyze blame* is used to print a list of all running units based on the time they took to initialise. This information is very helpful and can be used to optimise boot-up times. *systemd-analyze blame* doesn't display results for services with *Type=simple*, because *systemd* considers such services to be started immediately; hence, no measurement of the initialisation delays can be done.

Figure 4 shows the output of *systemd-analyze blame*.

The following command prints a tree of the time-critical chain of units:

```
command systemd-analyze critical-chain
```

Figure 5 shows the output of the command *systemd-analyze critical-chain*.

Steps to reduce the boot-up time

Shown below are the various steps that can be taken to reduce boot-up time.

BUM (Boot-Up-Manager): BUM is a run level configuration editor that allows the configuration of *init* services when the system boots up or reboots. It displays a list of every service that can be started at boot. The user can toggle individual services on and off. BUM has a very clean GUI and is very easy to use.

BUM can be installed in Ubuntu 14.04 using the following command:

```
sudo apt-get install bum
```

To install it in versions later than 15.10, download the packages from the

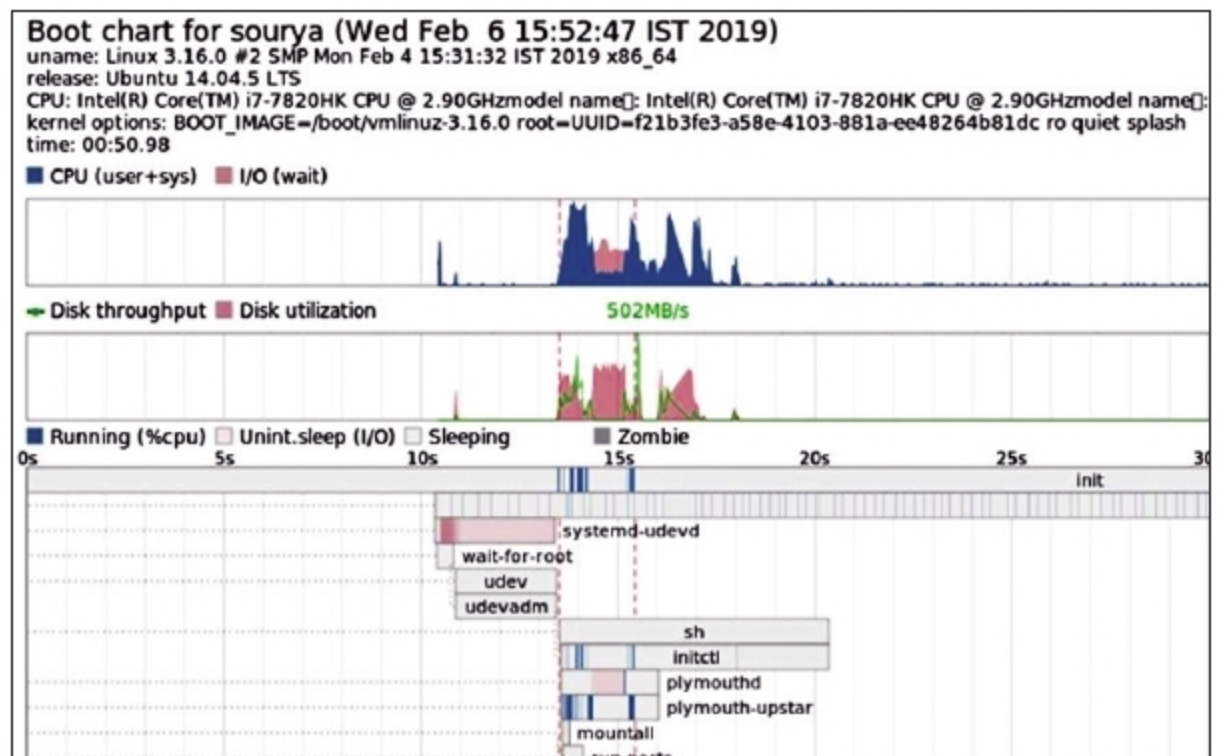


Figure 2: Boot chart

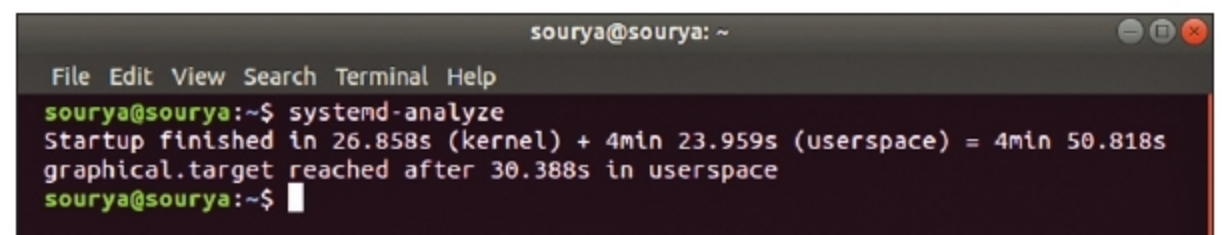


Figure 3: Output of *systemd-analyze*

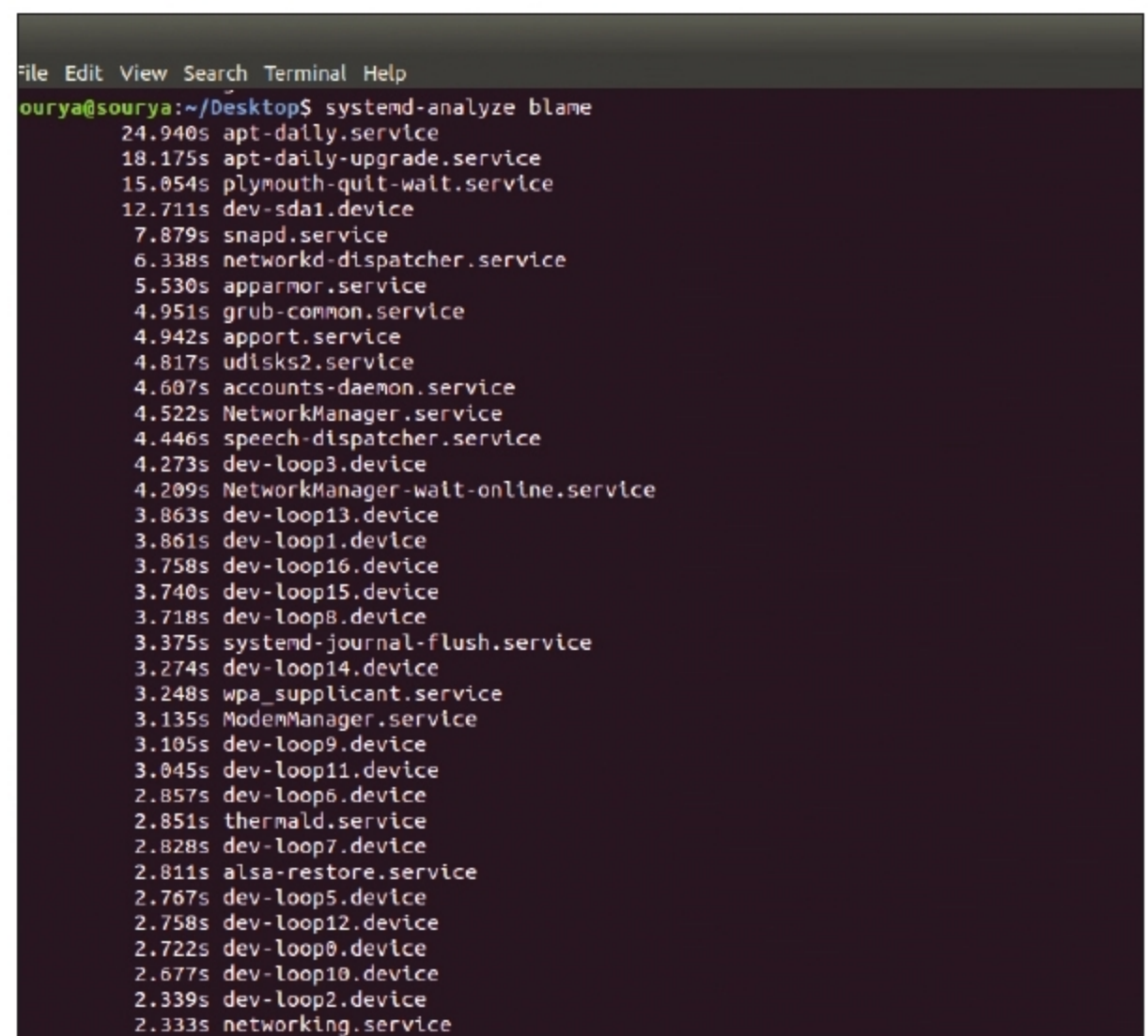


Figure 4: Output of *systemd-analyze blame*

link <http://apt.ubuntu.com/p/bum> 13.

Start with basic things and disable

services related to the scanner and printer. You can also disable Bluetooth